Do quasi-star photospheres survive infall?

A steady-state test of the MESA-QUEST "Little Red Dot" models

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Abstract

Santarelli, Bellinger et al. (2026, ApJL 998, L4) model Little Red Dots (LRDs) as late-stage quasi-stars: $\sim 10^6\,M_\odot$, fully convective envelopes around accreting black holes, with $L = L_{\rm Edd}(\kappa_{\rm es})$, $T_{\rm eff} \approx 6000\,{\rm K}$, $\log g \approx 0$. Their spectra come from LTE ATLAS9 atmospheres. We ask two questions. (i) Given their stated surface properties, is the photosphere plausibly in LTE? (ii) Does the $\sim 6000\,{\rm K}$ hydrostatic photosphere survive the external infall that the direct-collapse scenario requires? We build a quasi-static Z = 0 envelope model in Julia (Saha EOS, Rosseland continuum opacity, MLT, point-mass black hole) and replace the grey photosphere by a free-fall accretion shock. Without infall, the model reproduces their surface parameters to ∼15% in $T_{\rm eff}$. With infall we find two steady branches at every $\dot{M} > 0$ tested. One is an essentially unchanged "extended" star. The other is a "compact" star 2.5–55× smaller with a 16–60 kK boundary, buried in an infall that is optically thick even in LTE. The two branches approach each other and disappear (a fold) at $\dot{M} \approx 9$–$10\,M_\odot\,{\rm yr}^{-1}$ for $M_{\rm BH}/M = 0.1$ and 20–25 for 0.01 (for $\dot{M}/f$ if the infall arrives in streams covering a fraction f); above that no steady hydrostatic solution exists. The *observed* photosphere is at 5700–7600 K on both branches for $\dot{M} \geq 1\,M_\odot\,{\rm yr}^{-1}$ ($\approx 12\,{\rm kK}$ at $\dot{M} = 0.3$ on the compact branch). On the compact branch, however, it lies in infalling gas, and the LTE assumption is weakest exactly there. Simple rate estimates indicate that hydrogen at the paper's photosphere is strongly radiatively controlled, i.e. non-LTE.

Status labels. Throughout, [computed] means a result of the code described here; [estimate] means an order-of-magnitude rate or budget estimate with textbook approximations; [interpretation] means our reading of the results, not separately derived.

1 Context: the MESA-QUEST LRD models

The fiducial model of Santarelli et al. has $M = 10^6\,M_\odot$, $Z = 0$, $L \simeq 3.72\times10^{10}\,L_\odot = L_{\rm Edd}(\kappa_{\rm es})$, and at the onset of the late stage ($M_{\rm BH}/M \simeq 0.1$) $T_{\rm eff} = 6241\,{\rm K}$ and $\log g = -0.003$. These values imply $R = (L/4\pi\sigma T_{\rm eff}^4)^{1/2} = 1.15\times10^{16}\,{\rm cm}$ ($1.65\times10^5\,R_\odot$, 770 AU) and $M = gR^2/G = 0.99\times10^6\,M_\odot$, which is self-consistent. Their eq. (5) normalises $t_{\rm KH}$ to $R = 2\times10^4\,R_\odot$, a factor ≈ 8 smaller than the radius implied by their own L and $T_{\rm eff}$.

From the public repository (`andysantarelli/MESA-QUEST`, directory `qs_LRD_ApJL2026`) we read the following [verified by reading inlists/source]:

- `MLT_option='TDC'`, $\alpha_{\rm MLT} = 0.8$, `use_superad_reduction=.false.` (no MLT++).
- Default MESA grey T–τ outer boundary; no outer-boundary accretion.
- $L_{\rm BH} = \alpha\,4\pi cGM_\star/\kappa$ with κ taken in the innermost cell; $\alpha = 1$, $\epsilon = 0.1$.
- Inner boundary at the saturated-convection radius (tabulated, `Ki_tab.dat`) with a cavity mass added to the core.
- `convergence_ignore_equL_residuals=.true.` and an energy-conservation tolerance of 10^{99}.
- Synthetic spectra from the MESA `colors` module, nearest ATLAS9 grid point (LTE, metal-bearing; the paper notes $\log g$ lies outside the grid for most models).

2 Is the photosphere in LTE?

We estimate conditions at $\tau = 2/3$ for a pure H/He gas with $P = (2/3)g/\kappa$ and H^-, H bound-free and electron-scattering opacity, and compare collisional with radiative rates for hydrogen [estimate]. Collisional ionization from $n = 2$ uses a Seaton-type rate; photoionization from $n = 2$ uses $J_\nu = B_\nu(T_{\rm eff})$ and Kramers cross sections; collisional de-excitation of Hα uses a van Regemorter rate.

	quasi-star	same $T_{\rm eff}$, $\log g = 1$	same $T_{\rm eff}$, $\log g = 4.4$
n_e at $\tau = 2/3$ (cm^{-3})	5×10^{11}	1×10^{12}	1.6×10^{13}
H ionized fraction	0.025	0.011	8×10^{-4}
$n = 2$: $C_{\rm ion}/R_{\rm ion}$	2×10^{-3}	4×10^{-3}	5×10^{-2}
Hα: C_{32}/A_{32}	2×10^{-3}	4×10^{-3}	5×10^{-2}
$P_{\rm rad}/P_{\rm gas}$	0.2	0.04	2×10^{-4}

The $n = 2$ population, which controls the Balmer-continuum opacity and hence the Balmer break, is set by radiative rather than collisional processes by a factor ~ 500. Above the thermalisation depth one expects overionization of $n = 2$ relative to LTE. This is the regime in which NLTE departures in hydrogen are already documented for A–F supergiants at $\log g \approx 1$–2, and the quasi-star sits below that. The Balmer break, the feature the paper reproduces most closely (3.97 vs. 3.92 ± 0.03 for UNCOVER-45924), is therefore the least LTE-safe observable. These estimates do not tell us the sign or size of the NLTE change; that needs an NLTE calculation.

Related numbers [estimate]: escape speed $v_{\rm esc} = \sqrt{2gR} = \sqrt{2}(GMg)^{1/4} \approx 1500\,{\rm km\,s^{-1}}$. The strong-shock temperature for free-fall infall is $T_{\rm sh} = \frac{3}{16}\mu m_H v_{\rm ff}^2/k \approx 3 \times 10^7\,{\rm K}$. In LTE, H_2 is negligible at the photosphere: $n_{H_2}/n_{\rm H} \sim 4 \times 10^{-8}$ at 6200 K, rising only to $\sim 10^{-3}$ at 3000 K and becoming significant below ~ 2500 K at these densities.

3 Model

The code (`QuasiStarInfall.jl`) is quasi-static and spherically symmetric. A steady state is justified because $t_{\rm KH} = GM^2/(RL) \approx 5 \times 10^3\,{\rm yr} \ll M/\dot{M} \sim 10^6\,{\rm yr}$ at $1\,M_\odot\,{\rm yr}^{-1}$.

3.1 Equation of state

The composition is $X = 0.76$, $Y = 0.24$, $Z = 0$ (as in the MESA-QUEST inlist). We use an ideal gas of H, H^+, He, He^+, He^{2+}, e^- in LTE plus blackbody radiation. The Saha equations, with $\lambda_e^{-3} \equiv (2\pi m_e kT/h^2)^{3/2}$, are

$$\frac{n_{\rm H^+} n_e}{n_{\rm H^0}} = \lambda_e^{-3} e^{-\chi_{\rm H}/kT}, \qquad \frac{n_{\rm He^+} n_e}{n_{\rm He^0}} = 4\lambda_e^{-3} e^{-\chi_{\rm He\,I}/kT}, \qquad \frac{n_{\rm He^{2+}} n_e}{n_{\rm He^+}} = \lambda_e^{-3} e^{-\chi_{\rm He\,II}/kT}, \tag{1}$$

closed by charge neutrality (solved by bisection in $\ln n_e$). The pressure is $P = (n_{\rm H} + n_{\rm He} + n_e)kT + aT^4/3$. The specific entropy is the Sackur–Tetrode sum plus radiation,

$$s = \frac{k}{\rho}\sum_i n_i \left[\frac{5}{2} + \ln\!\left(\frac{g_i}{n_i}\Big(\frac{2\pi m_i kT}{h^2}\Big)^{3/2}\right)\right] + \frac{4aT^3}{3\rho}. \tag{2}$$

All thermodynamic derivatives are taken numerically in $(\ln\rho, \ln T)$, which stays well conditioned for $\beta \equiv P_{\rm gas}/P \ll 1$. With $\chi_\rho = \partial\ln P/\partial\ln\rho|_T$, $\chi_T = \partial\ln P/\partial\ln T|_\rho$, $s_\rho = \partial s/\partial\ln\rho|_T$ and $s_T = \partial s/\partial\ln T|_\rho$:

$$\nabla_{\rm ad} = \left(\chi_T - \chi_\rho \frac{s_T}{s_\rho}\right)^{-1}, \quad \Gamma_1 = \chi_\rho - \chi_T\frac{s_\rho}{s_T}, \quad \delta = \frac{\chi_T}{\chi_\rho}, \quad c_P = s_T - s_\rho\,\delta. \tag{3}$$

[computed] Checks: the fully ionized gas+radiation $\Gamma_1(\beta) = (32-24\beta-3\beta^2)/(24-21\beta)$ is reproduced to 10^{-5} for $\beta = 0.5, 0.1, 0.01$. The limits $\nabla_{\rm ad} \to 0.4$ (ideal gas) and 0.25 (radiation) are recovered. In the H ionization zone at $P_{\rm gas} \approx 30\,{\rm dyn\,cm^{-2}}$, Γ_1 drops to ≈ 0.97 at 8000 K.

3.2 Opacity

We use the Rosseland mean $\kappa_R^{-1} = \int (\kappa_\nu^{\rm abs} + \kappa_\nu^{\rm sc})^{-1} \partial_T B_\nu \, d\nu / \int \partial_T B_\nu \, d\nu$ on 4000 frequencies, from these continuum sources: Thomson scattering; H Rayleigh scattering; hydrogenic bound-free from H I ($n \le 12$) and He II ($n \le 8$) with LTE level populations and Kramers cross sections $\sigma_n = 2.815 \times 10^{29} Z^4/(n^5\nu^3)$; He I ground-state bound-free; free-free on H^+, He^+, He^{2+} ($\propto Z^2 n_e n_i T^{-1/2} \nu^{-3}$, Gaunt factor 1); and H^- bound-free and free-free (John 1988). Stimulated emission is included. **No bound–bound lines and no molecules** are included. κ_R is therefore a lower bound near 10^4–10^5 K, where line opacity in OPAL-type tables can raise it by a factor of a few. The table spans $\log T = 3.3$–8.0 and $\log R = -12$–1 ($R = \rho/T_6^3$). [computed] Checks: high-T limit $\kappa_R \to 0.350$ (vs. $0.2(1+X) = 0.352$ for m_p vs. m_H); H^- bound-free peak 4.0×10^{-17} cm^2 at $0.85\,\mu$m.

3.3 Envelope

The envelope is hydrostatic, with constant $L = L_{\rm Edd}(\kappa_{\rm es}, M)$ carried from the central source, and a point mass $M_{\rm BH}$ at the centre. We integrate inward in $x = \ln P$:

$$\frac{dr}{dx} = -\frac{P r^2}{G m \rho}, \qquad \frac{dm}{dx} = 4\pi r^2 \rho \frac{dr}{dx}, \qquad \frac{d\ln T}{dx} = \nabla, \tag{4}$$

with $\nabla_{\rm rad} = 3\kappa L P/(16\pi a c G m T^4)$. In convective regions ∇ follows from local MLT (Böhm-Vitense; Kippenhahn & Weigert eq. 7.18) with $\alpha_{\rm MLT} = 0.8$, $\ell = \alpha H_P$:

$$(\xi - U)^3 + \tfrac{8U}{9}\left(\xi^2 - U^2 - W\right) = 0, \quad U = \frac{3acT^3}{c_P \rho^2 \kappa \ell^2}\sqrt{\frac{8H_P}{g\delta}}, \quad W = \nabla_{\rm rad} - \nabla_{\rm ad}, \tag{5}$$

$$\nabla = \xi^2 - U^2 + \nabla_{\rm ad}, \qquad v_c^2 = \frac{g\delta\,(\xi - U)^2 \ell^2}{8H_P}. \tag{6}$$

The shooting condition is $m(r_{\rm in}) = M_{\rm BH}$ at $r_{\rm in} = \varepsilon_{\rm in} R$, with R found by scanning $\log R$ and bisecting every sign change. We use a Dormand–Prince 5(4) integrator with relative tolerance 10^{-7}. We report the local Eddington factor $\Gamma = \kappa L/(4\pi c G m)$ for radiation alone and the MLT Mach number v_c/c_s.

3.4 Outer boundary

No infall (equivalent to MESA's default): Eddington grey photosphere at $\tau = 2/3$,

$$T_b = T_{\rm eff}, \qquad P_b = \tfrac{2}{3}\Big(\frac{g}{\kappa_b} + \frac{F}{c}\Big) \iff P_{{\rm gas},b} = \tfrac{2}{3}\frac{g}{\kappa_b}\,(1 - \Gamma_\kappa), \qquad \Gamma_\kappa = \frac{\kappa_b F}{cg}. \tag{7}$$

Spherical free-fall infall onto a radiative shock at R. Upstream, $\rho_{\rm in}(r) = \dot M/(4\pi r^2 v_{\rm ff}) \propto r^{-3/2}$ with $v_{\rm ff} = (2GM/r)^{1/2}$. Momentum balance across the shock and settling layer gives

$$P_b = \rho_{\rm in} v_{\rm ff}^2 + \tfrac{2}{3}\frac{g}{\kappa_b} + \frac{F_{\rm tot}}{c}\Big(\tau_{\rm in} + \tfrac{2}{3}\Big), \qquad T_b^4 = \tfrac{3}{4}\,T_L^4\Big(\tau_{\rm in} + \tfrac{4}{3}\Big), \tag{8}$$

with $T_L^4 = L_{\rm tot}/(4\pi\sigma R^2)$ and $L_{\rm tot} = L + GM\dot M/R$. The infall optical depth $\tau_{\rm in} = \int_R^\infty \kappa_R \rho_{\rm in}\, dr$ uses the LTE opacity of the infall at the Lucy (1971) temperature $T^4(r) = T_L^4[W(r) + \tfrac{3}{4}\tau(r)]$ with $W = \tfrac{1}{2}[1 - (1 - R^2/r^2)^{1/2}]$, integrated inward from $10^4 R$. This boundary reduces exactly to the photospheric case for $\dot M \to 0$. If $\tau_{\rm in} > 2/3$, the observed photosphere is in the infall at $r_{\rm ph}$ where $\tau = 2/3$, and we quote $T_{\rm obs} = [L_{\rm tot}/(4\pi\sigma r_{\rm ph}^2)]^{1/4}$. A stream covering a fraction f of the surface at rate $\dot M$ has, at its footprint, the conditions of the spherical model at $\dot M/f$.

3.5 Infall diagnostics

[estimate]

- Post-shock cooling: bremsstrahlung, $\Lambda = 2.4\times10^{-27}T^{1/2}$, compression 4. The cooling length is $\ell_{\rm cool} = (v_{\rm ff}/4)\,3kT_{\rm sh}/(n\Lambda)$.
- Ionization budget: ionizations from half of $L_{\rm acc}$, at 35 eV (secondary-electron limit) or 1 keV per ionization, versus case-B recombinations in the infall out to $10R$: $\dot N_{\rm rec} = \alpha_B n_R^2\, 4\pi R^3 \ln 10$.
- Radiation force on ionized infall: $\Gamma_{\rm es} = L_{\rm tot}/L_{\rm Edd}(\kappa_{\rm es}, M) \geq 1$.

4 Results

4.1 Validation without infall

[computed]

$M_{\rm BH}/M$	$\varepsilon_{\rm in}$	R (cm)	$T_{\rm eff}$ (K)	$\log g$	$P_{{\rm gas},b}$	Γ_κ	max Mach
0.1	10^{-4}	8.30×10^{15}	7335	0.285	0.26	0.93	0.13
0.1	10^{-3}	8.47×10^{15}	7261	0.267	0.36	0.91	0.13
0.1	10^{-2}	8.73×10^{15}	7151	0.241	0.98	0.77	0.13
0.01	10^{-4}–10^{-2}	1.35×10^{16}	5753	−0.137	57	0.024	0.66
Santarelli et al.		1.15×10^{16}	6241	−0.003	–	–	–

The model brackets the published surface parameters. We get $T_{\rm eff}$ within $+16\%/-8\%$ and $\log g$ within 0.3 dex, and the inner-radius regularisation changes R by $< 5\%$. Differences from MESA-QUEST are expected: OPAL tables vs. our line-free opacity, TDC vs. MLT, and a tabulated saturated-convection inner boundary vs. a point mass. In the bulk of the envelope $\beta = 0.0060$–0.0075, consistent with the Eddington quartic $1-\beta = 0.003\,\mu^4\beta^4(M/M_\odot)^2$ ($\beta = 0.0073$ for $\mu = 0.588$).

Fig. 1 shows the local Eddington factor and MLT Mach number. For $M_{\rm BH}/M = 0.01$ the H-ionization peak is locally super-Eddington for radiation alone ($\Gamma \approx 9$ at $T \approx 9500$ K), carried by convection at Mach ≤ 0.66. For $M_{\rm BH}/M = 0.1$ the outer envelope sits at $\Gamma \approx 1$ with no ionization peak, because the density there is low. In both cases Γ rises deep inside only because L is fixed at $L_{\rm Edd}(M)$ while the enclosed mass $m < M$. Convection is subsonic everywhere in MLT. Note that for $M_{\rm BH}/M = 0.1$ the photosphere itself has $\Gamma_\kappa \approx 0.9$, so $P_{\rm gas} \approx 0.4\,{\rm dyn\,cm^{-2}}$ is much smaller than $P_{\rm rad} \approx 7\,{\rm dyn\,cm^{-2}}$.

4.2 With infall: two branches and a fold

[computed] Scanning $\log R \in [14, 16.6]$ in steps of 0.025 and bisecting every sign change gives the solutions in Table 1 and Fig. 2. (An earlier version of this analysis that kept only the first root reported a spurious jump at $\dot M \approx 3$; it is superseded.)

Why infall cannot make a bigger star. [interpretation, supported by the β values above] With $\beta \approx 0.007$ nearly constant, the envelope is close to an Eddington standard model, whose mass fixes β and hence the adiabat (specific entropy). At fixed M the adiabat is therefore pinned. The outer boundary only selects where the adiabat is truncated. Ram pressure raises P_b, so the truncation moves deeper, to a higher (P, T), and the surface becomes smaller and hotter. Accretion-bloated protostars (Hosokawa et al. 2012, 2013) swell for a different reason: entropy deposition in a gas-pressure-dominated star. That route is not available here.

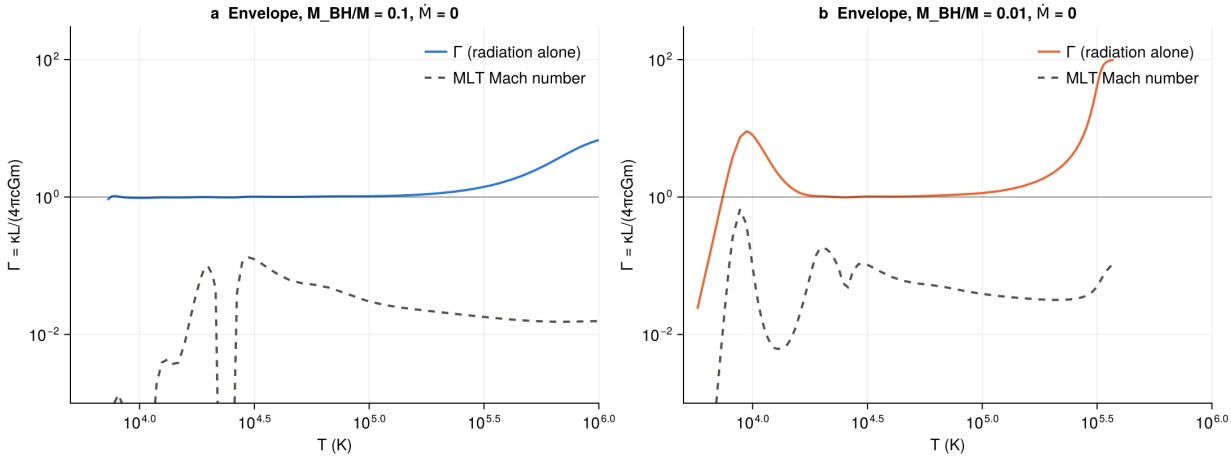


Figure 1: No-infall envelopes: local Eddington factor for radiation alone and MLT Mach number vs. temperature.

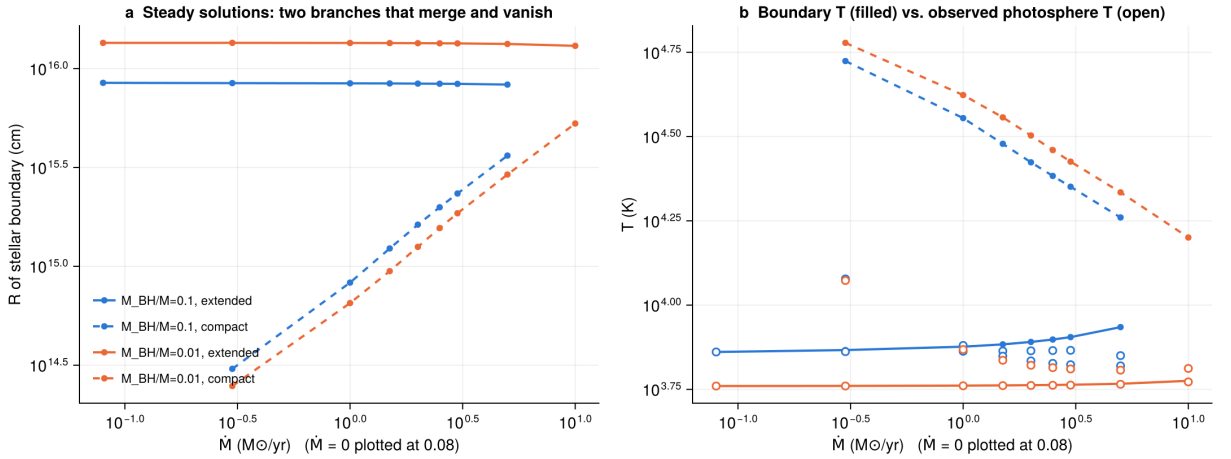


Figure 2: (a) Stellar boundary radius of all steady solutions vs. $\dot{M}$. (b) Boundary temperature (filled) and observed-photosphere temperature (open; equal to T_b when $\tau_{\rm in} < 2/3$).

What an observer sees. On the compact branch the photosphere lies in the infall at $r_{\rm ph} \approx 2$–$13\,R$, i.e. $r_{\rm ph} \approx 3 \times 10^{15}$–$1.1 \times 10^{16}$ cm, approaching the extended-star radius for $\dot{M} \geq 1$. There $T_{\rm obs} \approx 6400$–7600 K, rising to $\approx 1.2 \times 10^4$ K at $\dot{M} = 0.3$. [interpretation] The likely reason is that LTE hydrogen recombination makes the opacity collapse near 6–7 kK wherever the gas sits, so both branches show a 6000–7500 K surface for $\dot{M} \gtrsim 1$. On the compact branch, however, it is an infalling (several hundred km s^{-1}), X-ray-irradiated layer, not a hydrostatic photosphere. Lines should be redshifted, and LTE is least justified there.

Confined super-Eddington layers. At the boundary, $\Gamma_\kappa = \kappa_b F/(cg)$ exceeds unity on the compact branch (1.4–3.9) and on the $M_{\rm BH}/M = 0.1$ extended branch for $\dot{M} \geq 1$ (1.15–5.8). These layers are hydrostatic only because the infall's ram pressure holds them down, a configuration expected to be Rayleigh–Taylor unstable [interpretation].

4.3 Infall diagnostics on the extended branch

[estimate] For $M_{\rm BH}/M = 0.1$:

$M_{\rm BH}/M$	$\dot{M}$	branch	R (cm)	T_b (K)	$\tau_{\rm in}^{\rm LTE}$	$r_{\rm ph}/R$	$T_{\rm obs}$ (K)
0.1	0	–	8.47×10^{15}	7261	0	1	7261
0.1	0.3	compact	3.03×10^{14}	52976	3.24	10.6	11958
		extended	8.44×10^{15}	7356	0.06	1	7276
0.1	1	compact	8.27×10^{14}	35871	5.74	9.69	7592
		extended	8.42×10^{15}	7528	0.18	1	7294
0.1	3	compact	2.33×10^{15}	22450	7.28	4.48	6656
		extended	8.37×10^{15}	8036	0.58	1	7342
0.1	5	compact	3.63×10^{15}	18197	7.61	2.93	6602
		extended	8.30×10^{15}	8604	1.10	1.09	7082
0.1	≥ 10	no steady solution in 10^{14}–4×10^{16} cm					
0.01	0	–	1.35×10^{16}	5753	0	1	5753
0.01	0.3	compact	2.48×10^{14}	59989	3.65	13.3	11829
		extended	1.35×10^{16}	5758	0.003	1	5754
0.01	1	compact	6.52×10^{14}	41997	6.78	13.1	7382
		extended	1.35×10^{16}	5770	0.007	1	5763
0.01	3	compact	1.86×10^{15}	26648	9.26	6.03	6465
		extended	1.34×10^{16}	5804	0.01	1	5790
0.01	10	compact	5.28×10^{15}	15871	9.29	2.12	6489
		extended	1.30×10^{16}	5962	0.04	1	5918
0.01	≥ 25	no steady solution in 10^{14}–4×10^{16} cm					

Table 1: Steady solutions ($M = 10^6\,M_\odot$, $L = L_{\rm Edd}$). $\dot{M}$ in $M_\odot\,{\rm yr}^{-1}$ (or $\dot{M}/f$ for streams). Fold refinement ($\Delta\log R = 0.01$): for $M_{\rm BH}/M = 0.1$ the two roots are at $\log R = 15.825$ and 15.895 at $\dot{M} = 9$ and absent at 10; for 0.01 they are at 16.005 and 16.075 at $\dot{M} = 20$ and absent at 25.

$\dot{M}$	$\rho_{\rm in}v_{\rm ff}^2$ (dyn cm^{-2})	$\tau_{\rm in}^{\rm LTE}$	$\tau_{\rm in}^{\rm es}$	$\ell_{\rm cool}/R$	ion/rec (35 eV–1 keV)	$\Gamma_{\rm es}$
0.01	0.12	0.003	0.024	0.37	6.0–0.21	1.0001
0.03	0.37	0.008	0.071	0.12	2.0–0.07	1.0002
0.1	1.24	0.023	0.24	0.037	0.60–0.021	1.0007
0.3	3.74	0.060	0.71	0.012	0.20–0.007	1.0021
1	12.6	0.18	2.4	0.004	0.06–0.002	1.0070

These numbers raise two consistency issues that the steady model cannot resolve.

1. Because $L \geq L_{\rm Edd}(\kappa_{\rm es})$, *ionized* infall is weightless or pushed outward, so it cannot be in free fall. Free-fall infall must be largely neutral.
2. Whether the shock keeps the infall ionized depends on the energy per ionization. At low $\dot{M}$ it likely does; at high $\dot{M}$ recombinations ($\propto \dot{M}^2$) outrun ionizations ($\propto \dot{M}$). On the compact branch the gas near the shock is at 16–60 kK and ionized, so the free-fall boundary is internally inconsistent there. The thin-shock assumption fails for $\dot{M} \lesssim 0.03$ ($\ell_{\rm cool}/R \gtrsim 0.1$).

5 Limitations

- 1D, spherical, steady; local MLT (not TDC); no turbulent pressure; no rotation.
- LTE throughout, including the infall. Photoionization by the $\sim 3\times10^7$ K shock is ignored except in the budget estimate. The radiation force on the infall is ignored in the boundary condition.
- Continuum-only Rosseland opacity (no H/He lines, no molecules).
- Inner boundary is a point mass with $m(\varepsilon_{\rm in}R) = M_{\rm BH}$, not the MESA-QUEST saturated-convection radius.

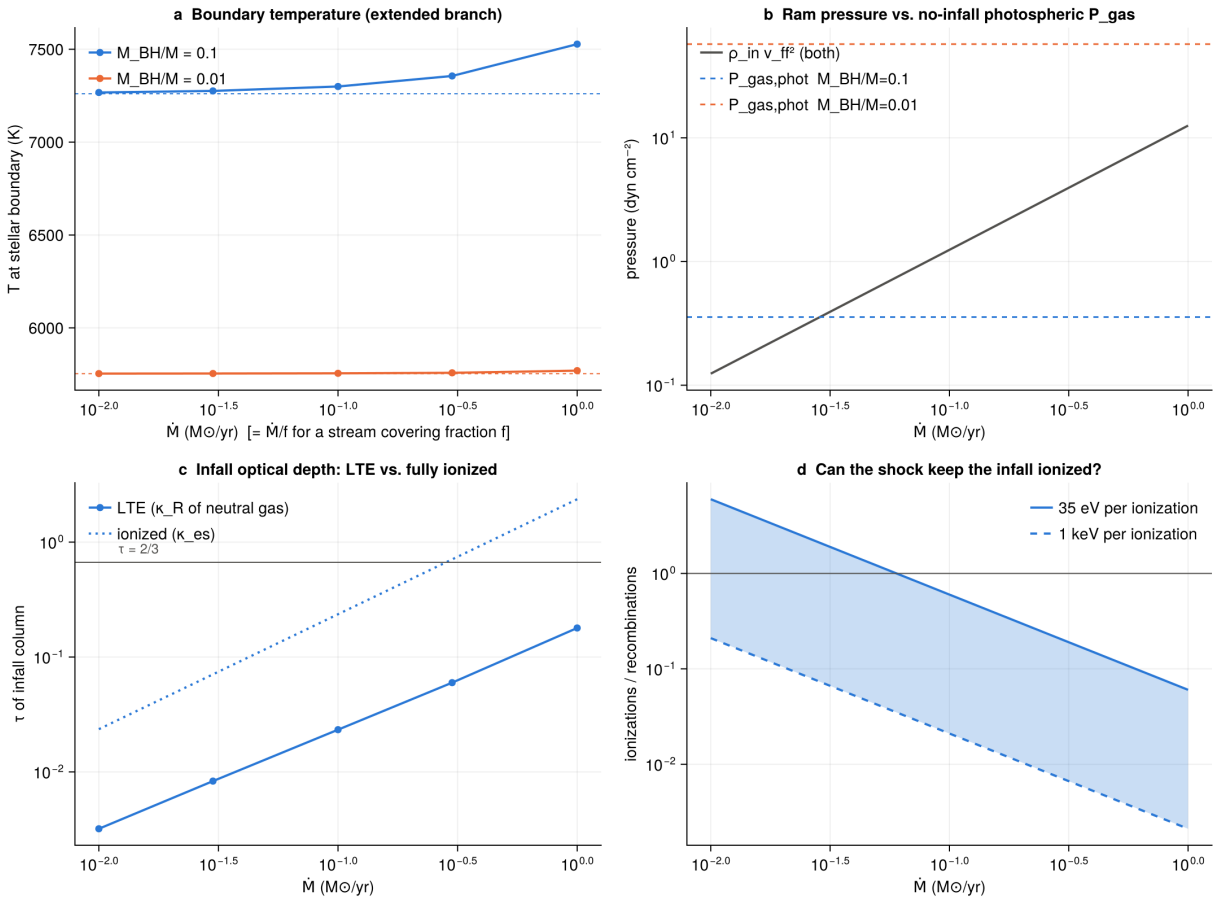


Figure 3: Extended-branch diagnostics vs. $\dot{M}$. (a) Boundary temperature. (b) Ram pressure vs. photospheric gas pressure without infall. (c) Infall optical depth, LTE vs. fully ionized. (d) Ratio of ionizations driven by the shock to recombinations in the infall ($\leq 10R$), bracketed by 35 eV and 1 keV per ionization.

- Stability of the two branches has not been analysed. Which one nature selects, and what happens past the fold, requires dynamics.
- The §2 rate comparisons use textbook approximations and are good to a factor of a few.

6 Next steps

1. Linear stability of both branches. This is cheap and tells the dynamical runs what to look for.
2. Dynamical radiation hydrodynamics: TITAN (Gehmeyr & Mihalas 1994). The 1995 Fortran release now builds with gfortran; all benchmarks run, and the six with a fixed end time reproduce the 1995 step counts exactly. It runs as a shared library driven from Julia, bitwise identical to the standalone code. A Julia port (tolerance-verified, not bitwise) is under way. TITAN has no convection; a time-dependent convection model must be added.
3. Time-independent NLTE post-processing (H with 10–15 levels plus continuum, then He) on steady and dynamical snapshots. This is justified because recombination times (~ 10 s at the photosphere, $\lesssim 1$ d in the infall) are $\ll R/v_{\rm ff} \sim 2$ yr. The first target is the Balmer break on both branches.

Code and data

`QuasiStarInfall.jl` (Julia 1.11): `src/` (EOS, opacity, envelope, boundary) and `scripts/` (`build_kap.jl`, `validate2.jl`, `run_grid.jl`, `branches.jl`, `fold.jl`, `plot_grid.jl`). Results are in `output/*.csv`.

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